

Kernel-Perfect Orientations of Line Graphs of Complete Graphs

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Abstract

The list-edge-colouring conjecture asserts that $\chi(H) = \chi_\ell(H)$ for every line graph H . Galvin proved it for line graphs of bipartite graphs, using the kernel lemma of Bondy, Boppana and Siegel. We prove that for every even $m \geq 4$ the line graph $L(K_m)$ admits no kernel-perfect orientation whose out-degrees are all at most $\chi'(K_m) - 1 = m - 2$. Consequently the kernel lemma cannot establish $\chi'_\ell(K_m) = \chi'(K_m)$ for any even m , which is precisely the case of the conjecture that remains open for complete graphs. The proof rests on an observation we call the star lemma: in a kernel-perfect orientation of a line graph, the edges of G incident with a fixed vertex are linearly ordered. This converts the question into a numerical one about those orders, and a counting argument then closes it. The result strengthens a theorem of McDonald, who ruled out the same conclusion for the restricted family of Galvin orientations. All assertions are proved from first principles, and every numerical claim has been verified by exhaustive computation in the stated range.

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1 Introduction

All graphs are finite and simple unless stated otherwise. For a graph G we write $V(G)$ and $E(G)$ for its vertex and edge sets, $d_G(v)$ for the degree of v , and $\Delta(G)$ for the maximum degree. The *line graph* $L(G)$ has vertex set $E(G)$, two edges of G being adjacent in $L(G)$ exactly when they share an end.

A *list assignment* for a graph H gives each vertex v a set $C(v)$ of colours. An *L -colouring* is a proper colouring c with $c(v) \in C(v)$ for all v . The *list chromatic number* $\chi_\ell(H)$ is the least k

such that an L -colouring exists whenever $|C(v)| = k$ for every v . Since ordinary colouring is the case of equal lists, $\chi(H) \leq \chi_\ell(H)$ always, and the gap can be arbitrarily large in general.

Conjecture 1.1 (list-edge-colouring conjecture). $\chi(H) = \chi_\ell(H)$ for every line graph H .

Equivalently, $\chi'_\ell(G) = \chi'(G)$ for every multigraph G , where χ' is the chromatic index and χ'_ℓ the list chromatic index. The conjecture has been attributed independently to several authors and has been open since the 1970s. Galvin proved it for bipartite multigraphs, thereby settling the Dinitz problem. The tool is the following lemma.

Definition 1.2 (kernel). Let D be a digraph. A set $K \subseteq V(D)$ is a *kernel* of D if

- (i) K is independent in the underlying graph of D , and
- (ii) every $u \notin K$ has an arc $u \rightarrow w$ with $w \in K$.

D is *kernel-perfect* if every induced subdigraph of D has a kernel.

Lemma 1.3 (Bondy, Boppana and Siegel). *Let H be a graph and let D be an orientation of H which is kernel-perfect. If $|C(v)| \geq d_D^+(v) + 1$ for every v , then H has an L -colouring. In particular $\chi_\ell(H) \leq \Delta^+(D) + 1$.*

The following notion isolates exactly what Lemma 1.3 asks of a line graph.

Definition 1.4 (k -kernel-orientable). For an integer $k \geq 1$, a graph H is *k -kernel-orientable* if it has a kernel-perfect orientation D with $d_D^+(x) \leq k - 1$ for every $x \in V(H)$.

By Lemma 1.3, if $L(G)$ is $\chi'(G)$ -kernel-orientable then $\chi'_\ell(G) = \chi'(G)$, so conjecture 1.1 holds for G . Galvin's theorem is exactly the statement that $L(G)$ is $\chi'(G)$ -kernel-orientable whenever G is bipartite. Our main result says that this route is unavailable for complete graphs of even order.

Theorem 1.5. *Let $m \geq 4$ be even. Then $L(K_m)$ is not $(m - 1)$ -kernel-orientable. Since $\chi'(K_m) = m - 1$, the kernel lemma Lemma 1.3 cannot establish $\chi'_\ell(K_m) = \chi'(K_m)$ for any even m .*

The restriction to even m is not a weakness of the method but the point of the result. For odd m one has $\chi'(K_m) = m$, and Haggkvist and Janssen proved $\chi'_\ell(K_m) \leq m$, so conjecture 1.1 is settled for odd complete graphs. The even case is the one that remains open, and Theorem 1.5 says the chapter's own method is not available there.

Theorem 1.5 strengthens a theorem of McDonald, who proved the same conclusion for the restricted family of *Galvin orientations*. We discuss the relationship in section 7.

2 The star lemma

Throughout this section G is a graph and D is an orientation of $L(G)$. For $v \in V(G)$ let $E(v)$ denote the set of edges of G incident with v . Since any two members of $E(v)$ share the end v , the set $E(v)$ is a clique of $L(G)$, so D induces a tournament on it.

Lemma 2.1 (star lemma). *Let D be a kernel-perfect orientation of $L(G)$. Then for every $v \in V(G)$ the tournament induced by D on $E(v)$ is transitive.*

Proof. **Step 1.** Let T be a tournament and let K be a kernel of T . Since K is independent and any two vertices of a tournament are adjacent, $|K| \leq 1$; and $K \neq \emptyset$, since otherwise condition (ii) of Definition 1.2 would ask a vertex to have an arc into the empty set. So $K = \{x\}$ for a single vertex x , and (ii) says every other vertex has an arc to x . Hence x has out-degree 0 in T , that is, x is a sink.

Step 2. A directed triangle has no sink, since each of its three vertices has out-degree 1. By Step 1 it therefore has no kernel.

Step 3. $E(v)$ is a clique of $L(G)$, so every $S \subseteq E(v)$ induces a tournament in D . As D is kernel-perfect, that tournament has a kernel. By Step 2 no such S induces a directed triangle, so D has no directed triangle inside $E(v)$.

Step 4. A tournament with no directed triangle is transitive. Indeed, suppose $a \rightarrow b$ and $b \rightarrow c$ with $a \neq c$. The arc between a and c cannot be $c \rightarrow a$, for then $a \rightarrow b \rightarrow c \rightarrow a$ would be a directed triangle. Hence $a \rightarrow c$. $\square$

Remark 2.2. Lemma 2.1 is not new in content. Borodin, Kostochka and Woodall proved that an orientation of a line graph is kernel-perfect if and only if every clique has a kernel and every directed odd cycle has a chord or a pseudochord. The necessity of the clique condition, applied to every subset of $E(v)$, yields Lemma 2.1 at once. We have given the direct argument above so that this article remains self-contained, and because the formulation as a linear order, rather than as the existence of a kernel, is what makes Definition 2.4 available.

Remark 2.3. Galvin orients $L(K_{n,n})$ by the entries of a Latin square, running along each row from the smaller entry to the larger and along each column in the opposite direction. Rows and columns are precisely the sets $E(v)$ for $K_{n,n}$, so his construction is a linear order on each $E(v)$. Lemma 2.1 says this shape is forced: no other kind of orientation can be kernel-perfect.

Lemma 2.1 lets us replace an orientation by a family of numbers.

Definition 2.4 (the rank function). Let D be a kernel-perfect orientation of $L(G)$. For $v \in V(G)$ and $e \in E(v)$ put

$$M(v, e) = |\{f \in E(v) : e \rightarrow f \text{ in } D\}|.$$

For an edge $e = uv$ of G we abbreviate $M(u, v) := M(u, e)$ and $M(v, u) := M(v, e)$.

Lemma 2.5. *Let D be a kernel-perfect orientation of $L(G)$ and let $v \in V(G)$. Then*

- (a) $M(v, \cdot)$ is a bijection from $E(v)$ onto $\{0, 1, \dots, d_G(v) - 1\}$;
- (b) for $e, f \in E(v)$ with $e \neq f$ we have $e \rightarrow f$ in D if and only if $M(v, e) > M(v, f)$;
- (c) for every edge $e = uv$ of G , $d_D^+(e) = M(u, v) + M(v, u)$.

Proof. **Step 1.** By Lemma 2.1 the tournament on $E(v)$ is transitive, so it is a linear order: write $E(v) = \{e_1, \dots, e_d\}$ with $e_i \rightarrow e_j$ exactly when $i < j$, where $d = d_G(v)$. Then $M(v, e_i) = |\{j : j > i\}| = d - i$, which runs over $\{0, 1, \dots, d - 1\}$ bijectively as i runs from 1 to d . This proves (a), and (b) follows since $i < j$ is equivalent to $d - i > d - j$.

Step 2. The neighbours of $e = uv$ in $L(G)$ are the edges of G sharing u with e and the edges sharing v with e . As G is simple, no edge other than e lies in both $E(u)$ and $E(v)$, so these two sets of neighbours are disjoint. The out-neighbours of e therefore split accordingly, giving $d_D^+(e) = M(u, e) + M(v, e)$, which is (c). $\square$

3 The budget is tight on regular class one graphs

Recall that G is *class one* if $\chi'(G) = \Delta(G)$.

Lemma 3.1. *Let G be Δ -regular and class one, and suppose $L(G)$ is $\chi'(G)$ -kernel-orientable, witnessed by D . Then*

$$M(u, v) + M(v, u) = \Delta - 1 \quad \text{for every edge } uv \text{ of } G.$$

Proof. **Step 1.** Let $n = |V(G)|$, so $|E(G)| = n\Delta/2$. Two edges of G are adjacent in $L(G)$ exactly when they share an end, and by simplicity they share at most one end, so each vertex v contributes exactly $\binom{d_G(v)}{2}$ edges to $L(G)$ and no edge of $L(G)$ is counted twice. Hence

$$|E(L(G))| = \sum_{v \in V(G)} \binom{d_G(v)}{2} = n \binom{\Delta}{2} = \frac{n\Delta(\Delta-1)}{2}.$$

Step 2. Every edge of $L(G)$ receives exactly one orientation in D , so $\sum_{e \in E(G)} d_D^+(e) = |E(L(G))|$.

Step 3. By hypothesis $d_D^+(e) \leq \chi'(G) - 1 = \Delta - 1$ for every e , and there are $|E(G)| = n\Delta/2$ edges, so

$$\sum_{e \in E(G)} d_D^+(e) \leq (\Delta - 1) \cdot \frac{n\Delta}{2} = \frac{n\Delta(\Delta-1)}{2}.$$

Step 4. Steps 1 to 3 give equality throughout, so $d_D^+(e) = \Delta - 1$ for every edge e . Lemma 2.5(c) now gives the claim. $\square$

4 Proof of the main theorem

Proof of Theorem 1.5. Let $m \geq 4$ be even, write $G = K_m$, and put $s = m - 2$. Then G is $(m-1)$ -regular and $\chi'(G) = m - 1$, so G is class one with $\Delta = m - 1$ and $s = \Delta - 1$.

Suppose for a contradiction that $L(K_m)$ is $(m-1)$ -kernel-orientable, witnessed by a kernel-perfect orientation D with all out-degrees at most $m - 2$. Let M be the rank function of Definition 2.4.

Step 1. By Lemma 3.1,

$$M(u, v) + M(v, u) = s \quad \text{for every edge } uv. \quad (4.1)$$

By Lemma 2.5(a), each $M(u, \cdot)$ is a bijection onto $\{0, 1, \dots, s\}$, since $d_G(u) = m - 1 = s + 1$.

Step 2. Fix any vertex i of K_m . By Lemma 2.5(a) there is a unique neighbour j of i with $M(i, j) = 0$, and then (4.1) gives $M(j, i) = s$.

Step 3. Let k be any vertex other than i and j ; there are exactly $m - 2 = s$ such k . Since K_m is complete, ij , jk and ik are edges of K_m and hence vertices of $L(K_m)$, pairwise adjacent, so they induce a triangle in $L(K_m)$. We determine its orientation.

Step 4. *The arc between ij and jk .* These two share the end j , so by Lemma 2.5(b), $ij \rightarrow jk$ if and only if $M(j, i) > M(j, k)$. Now $M(j, i) = s$ is the largest value in the range of $M(j, \cdot)$, and $M(j, k) \neq M(j, i)$ because $M(j, \cdot)$ is injective and $k \neq i$. Hence $M(j, i) > M(j, k)$ and

$$ij \longrightarrow jk.$$

Step 5. *The arc between ik and ij .* These share the end i , so $ik \rightarrow ij$ if and only if $M(i, k) > M(i, j) = 0$. Since $M(i, \cdot)$ is injective and $k \neq j$, we have $M(i, k) \neq 0$, hence $M(i, k) > 0$ and

$$ik \longrightarrow ij.$$

Step 6. *The arc between jk and ik .* These share the end k , so $jk \rightarrow ik$ if and only if $M(k, j) > M(k, i)$. By (4.1), $M(k, j) = s - M(j, k)$ and $M(k, i) = s - M(i, k)$, so this condition is equivalent to $M(i, k) > M(j, k)$.

Step 7. If $M(i, k) > M(j, k)$ then, combining Steps 4, 5 and 6, we obtain

$$ij \longrightarrow jk \longrightarrow ik \longrightarrow ij,$$

a directed triangle in D . By Step 2 of the proof of Lemma 2.1 a directed triangle has no kernel, and it is an induced subdigraph of D , contradicting kernel-perfectness. Hence $M(i, k) \leq M(j, k)$. Moreover $M(k, i) \neq M(k, j)$ because $M(k, \cdot)$ is injective and $i \neq j$, so by (4.1) $M(i, k) \neq M(j, k)$. Therefore

$$M(i, k) < M(j, k) \quad \text{for every } k \notin \{i, j\}. \quad (4.2)$$

Step 8. Sum (4.2) over the s vertices $k \notin \{i, j\}$. Since $M(i, \cdot)$ is a bijection onto $\{0, 1, \dots, s\}$ and omits only the value $M(i, j) = 0$ when k ranges over $V \setminus \{i, j\}$,

$$\sum_{k \notin \{i, j\}} M(i, k) = \left(\sum_{t=0}^s t \right) - M(i, j) = \frac{s(s+1)}{2} - 0 = \frac{s(s+1)}{2}.$$

Likewise $M(j, \cdot)$ omits only the value $M(j, i) = s$, so

$$\sum_{k \notin \{i, j\}} M(j, k) = \frac{s(s+1)}{2} - s.$$

Step 9. Summing (4.2) over all $s \geq 1$ choices of k gives a strict inequality:

$$\frac{s(s+1)}{2} < \frac{s(s+1)}{2} - s,$$

that is $0 < -s$. Since $s = m - 2 \geq 2$ this is false. The assumed orientation cannot exist. $\square$

Remark 4.1. The argument never analyses a triangle in isolation. Proposition 5.1 below shows that for $\Delta \geq 4$ a single triangle imposes no contradiction at all, so no local argument can prove Theorem 1.5. The force comes from choosing j at the extreme of the order at i , which fixes two of the three arcs of *every* triangle through ij at once and makes all of them yield the same inequality (4.2). The contradiction is then a statement about row sums.

5 Sharpness and scope

Two remarks delimit Theorem 1.5. The first shows that at $\Delta = 3$ a purely local argument does suffice, and the second shows that it does not for larger Δ .

Proposition 5.1. *Let G be 3-regular and class one. If G contains a triangle, then $L(G)$ is not 3-kernel-orientable.*

Proof. Suppose D witnesses 3-kernel-orientability. By Lemma 3.1 with $\Delta = 3$ we have $s = 2$ and $M(u, v) + M(v, u) = 2$ for every edge. Let ijk be a triangle of G and set $p = M(i, j)$, $q = M(j, k)$ and $r = M(i, k)$, so that $M(j, i) = 2 - p$, $M(k, j) = 2 - q$ and $M(k, i) = 2 - r$. Injectivity of $M(i, \cdot)$, $M(j, \cdot)$ and $M(k, \cdot)$ gives respectively

$$p \neq r, \quad p + q \neq 2, \quad q \neq r.$$

By Lemma 2.5(b), as in Steps 4 to 6 above, the triangle ij, jk, ik of $L(G)$ is directed precisely when

$$(p + q < 2 \text{ and } r > p \text{ and } r > q) \quad \text{or} \quad (p + q > 2 \text{ and } r < p \text{ and } r < q).$$

There are nine pairs $(p, q) \in \{0, 1, 2\}^2$. The three with $p + q = 2$ are excluded. For the remaining six, avoiding a directed triangle forces r into a value already excluded:

(p, q)	requirement for no directed triangle	conflict
$(0, 0)$	$r \leq 0$	contradicts $r \neq p = 0$
$(0, 1)$	$r \leq 1$, and $r \neq 0$, so $r = 1$	contradicts $r \neq q = 1$
$(1, 0)$	$r \neq 2$, and $r \neq 1$, so $r = 0$	contradicts $r \neq q = 0$
$(1, 2)$	$r \neq 0$, and $r \neq 1$, so $r = 2$	contradicts $r \neq q = 2$
$(2, 1)$	$r \neq 0$, and $r \neq 2$, so $r = 1$	contradicts $r \neq q = 1$
$(2, 2)$	$r = 2$	contradicts $r \neq p = 2$

Every case fails, so the triangle of $L(G)$ is directed and has no kernel. $\square$

Remark 5.2. Proposition 5.1 does not extend to larger degrees, in two distinct senses. First, the local obstruction disappears: for $s = \Delta - 1 \geq 3$ there are triples (p, q, r) satisfying the three injectivity conditions and yielding an undirected triangle, for instance $(p, q, r) = (0, 2, 1)$. Exhaustive enumeration gives 0, 0, 6, 18, 42, 78, 132 admissible triples for $s = 1, 2, 3, 4, 5, 6, 7$ respectively. Second, the presence of a triangle is not by itself an obstruction: the circulant $C_8(1, 2)$ is 4-regular and class one and contains eight triangles, yet admits a system of orders satisfying (4.1) with no directed triangle.

Remark 5.3. The proof of Theorem 1.5 uses only that i and j are adjacent and that every other vertex is adjacent to both. For a Δ -regular class one graph this says $N(i) \setminus \{j\} \subseteq N(j)$, and since both sets have size $\Delta - 1$ it forces $N[i] = N[j]$. Because j is determined by D rather than chosen, the argument as stated applies exactly to disjoint unions of complete graphs of even order. Extending it further requires a different way of aggregating the inequalities (4.2).

6 Computational verification

Every numerical assertion above has been checked independently.

- (1) K_4 , by exhaustive orientation search. $L(K_4)$ is the octahedron, with 6 vertices and 12 edges. All $2^{12} = 4096$ orientations were enumerated; the 38 with all out-degrees at most 2 were tested for kernel-perfectness by examining every one of the $2^6 - 1$ induced subdigraphs. Every one of the 38 contains a directed triangle, so none is kernel-perfect. This confirms Theorem 1.5 for $m = 4$ without using Lemma 2.1.
- (2) K_4 , K_6 and K_8 , by exhaustive search over rank functions. Backtracking over all M satisfying (4.1), injectivity of every row, and absence of a directed triangle finds no solution in each case, agreeing with Theorem 1.5.
- (3) *Lemma 2.1.* For every tournament on at most 6 vertices, the properties “every non-empty subset contains a sink” and “transitive” select the same tournaments, namely 2, 8, 64, 1024 and 32768 of them on 2, 3, 4, 5, 6 vertices.
- (4) *Remark 5.2.* The counts 0, 0, 6, 18, 42, 78, 132 and the example $C_8(1, 2)$ were produced by direct enumeration.

7 Relation to McDonald’s theorem

McDonald defines a *Galvin orientation* of $L(G)$ from two pieces of data: a partition of $V(G)$ into sets U and D , and a single k -edge-colouring φ of G . For adjacent $e_1, e_2 \in E(G)$ with $\varphi(e_1) < \varphi(e_2)$ sharing a vertex w , the edge of $L(G)$ between them is oriented from e_2 to e_1 if $w \in D$, and from e_1 to e_2 if $w \in U$. Such an orientation is *proper with respect to k* if it is kernel-perfect and has all out-degrees at most $k - 1$. She proves:

Theorem 7.1 (McDonald). $L(K_n)$ has no proper Galvin orientation with respect to $\chi'(K_n)$, for any $n \geq 4$.

Theorem 1.5 is a strengthening of Theorem 7.1 in the even case, for the following reason. In a Galvin orientation every vertex orders its incident edges by the *same* colouring φ , choosing only whether to read it upwards or downwards. By Lemma 2.1, a general kernel-perfect orientation gives each vertex an arbitrary linear order on its incident edges. To recover a single global colouring from a rank function M one would need to set $\varphi(uv) = M(v, u) + 1$ consistently, which requires each vertex to play the same role for all of its edges, that is, requires a partition of $V(G)$ with one end of every edge in each part. That is precisely bipartiteness. For non-bipartite G , and in particular for K_m , the rank functions covered by Theorem 1.5 therefore form a strictly larger family than the Galvin orientations covered by Theorem 7.1.

Informally, Theorem 7.1 says that Galvin’s construction does not reach complete graphs, while Theorem 1.5 says that the kernel lemma does not, however the orientation is built.

McDonald also proves a positive companion result: $L(K_{\Delta+1})$ does admit a proper Galvin orientation with respect to roughly $\lfloor 3\Delta/2 \rfloor$, which is far from the bound $\chi'_\ell(K_n) \leq n$ of Haggkvist and Janssen.

Remark 7.2. The two places where Theorem 1.5 would most plausibly already appear have been consulted. The article *On kernel-perfect orientations of line graphs* by Borodin, Kostochka and Woodall, Discrete Math. **191** (1998) 45–49, proves only the characterisation quoted in Remark 2.2; its statement concerns cliques and odd cycles in general and makes no assertion about complete graphs or about out-degree bounds. McDonald’s article proves Theorem 7.1, which as explained above concerns the strictly smaller family of Galvin orientations. Neither contains Theorem 1.5.

This is evidence and not proof. The author has not read the full text of the 1998 article, only its abstract and the statement as quoted by McDonald, and a search of the literature cannot establish a negative. Theorem 1.5 should be regarded as a strengthening of Theorem 7.1 whose novelty is likely but unconfirmed.

8 Concluding remarks

Theorem 1.5 closes the kernel route to conjecture 1.1 for complete graphs of even order, which is exactly where the conjecture remains open for this family. It is consistent with the shape of the literature: the positive results on complete graphs, such as those of Haggkvist and Janssen, of Schauz for K_{p+1} with p an odd prime, and the recent families K_{p-1} and K_{2p} , all proceed through the Alon and Tarsi polynomial method and its formulation in terms of signed one-factorizations, rather than through kernels.

It is worth recording that the two methods are incomparable rather than nested. The Alon and Tarsi method proves $\chi'_\ell(K_4) = 3$, which by Theorem 1.5 the kernel lemma cannot. In the other direction, Galvin’s orientation of $L(K_{3,3})$ is kernel-perfect with all out-degrees equal to 2, so the kernel lemma proves $\chi'_\ell(K_{3,3}) = 3$, whereas the Alon and Tarsi certificate for that orientation vanishes and the Alon and Tarsi number of $L(K_{3,3})$ is 4. Neither method subsumes the other.